

Thermodynamics – Assignment 02

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1 Introduction

This study evaluates the thermodynamic performance of a regenerative Rankine cycle designed to provide 500 kW of electric power to O’Bleness Hospital in Athens, Ohio. The proposed system incorporates a high-pressure steam turbine, a low-pressure steam turbine, a vacuum condenser, an open feedwater heater (OFWH), and two feed pumps. Steam is generated at 40 bar and 450°C using natural gas combustion and expanded in two stages, with controlled steam extraction to improve cycle efficiency through regenerative heating.

The analysis includes determination of steam flow rate, bleed fraction, thermal efficiency, second-law efficiency, lost work in each component, and maximum theoretical (Carnot) efficiency. The results provide insight into the performance benefits of regeneration and the thermodynamic behavior of a methane-fired Rankine power plant.

2 Background/Assumptions

2.1 Rankine Cycle Overview

The Rankine cycle is the fundamental thermodynamic cycle used in steam power plants. It converts thermal energy into mechanical work through the vaporization and expansion of water/steam in a turbine. A conventional Rankine cycle consists of four primary components:

1. Boiler (heat addition)
2. Turbine (expansion and work output)
3. Condenser (heat rejection)
4. Pump (pressurization of liquid)

In this design, a **regenerative Rankine cycle** is used. Regeneration improves efficiency by extracting steam from the turbine at an intermediate pressure and using it to preheat the feedwater in an open feedwater heater (OFWH). This reduces the heat input required in the boiler and improves overall cycle performance.

2.2 System Description

The proposed plant produces 500 kW of net electric power. The working fluid is water/steam.

Boiler (F-100)

- Steam conditions: 40 bar and 450°C
- Fuel: Natural gas (100% methane)
- High heating value (HHV): 55.5 MJ/kg
- Boiler efficiency: 85%

High-Pressure Turbine (ST-100)

- Inlet: 40 bar, 450°C
- Exit pressure: 2 bar
- Turbine efficiency: 85%
- A fraction of the steam is extracted at 2 bar for regeneration

Low-Pressure Turbine (ST-101)

- Inlet: 2 bar (remaining steam after extraction)
- Expands to condenser pressure
- Turbine efficiency: 85%

Condenser (E-100)

- Condenses steam to saturated liquid at 40°C
- Cooling water inlet temperature: 20°C
- Cooling water temperature rise: 10°C
- Heat sink temperature: 300 K

Pumps

- Condensate pump (P-100): raises pressure to 2 bar
- Feedwater pump (P-101): raises pressure to 40 bar
- Pump efficiency: 75%

2.3 Regeneration via Open Feedwater Heater

The open feedwater heater (OFWH) mixes:

- Extracted steam from the high-pressure turbine at 2 bar
- Condensate from the condenser after being pumped to 2 bar

The outlet condition of the OFWH is specified as 2 bar and 1°C subcooled liquid. The bleed fraction is determined from an energy balance around the OFWH. This regeneration increases the average temperature of heat addition in the boiler and improves cycle efficiency.

2.4 Fuel and Combustion Data

The fuel used in the boiler is 100% methane (CH_4). The combustion air composition is:

- 21 mol% O_2
- 79 mol% N_2

The dry flue gas contains 3 mol% O_2 , which allows determination of the excess air factor. The plant operates 8000 hours per year.

2.5 CHE calculator

The CHE calculator and its built in functions will be utilized to evaluate and optimize the problem.

3 Spreadsheet Setup and Property Calculations

The Excel model was structured by listing each thermodynamic stream in rows and organizing the associated thermodynamic properties in columns. For each stream, both actual and ideal (isentropic) states were recorded where necessary to compute component efficiencies.

3.1 Steam Property Functions

For vapor-phase properties, the following **CHECalculator** steam functions were used with the correctly corresponding cells in the same column:

$$h_v = \text{HSteam}(P, T, "v")$$

For liquid-phase properties:

$$h_l = \text{HSteam}(P, T, "l")$$

Similarly, entropy values were calculated using:

$$s_v = \text{SSteam}(P, T, "v")$$

$$s_l = \text{SSteam}(P, T, "l")$$

where:

- P = pressure (bar)
- T = temperature ($^{\circ}\text{C}$)
- “v” specifies vapor phase
- “l” specifies liquid phase

3.2 Two-Phase Mixture Calculations (Lever Rule)

For streams in the two-phase region, the thermodynamic properties were determined using the lever rule.

For any property ϕ (enthalpy or entropy):

$$\phi = (1 - x)\phi_f + x\phi_g$$

where:

- x = vapor quality
- ϕ_f = saturated liquid property
- ϕ_g = saturated vapor property

The molar enthalpy and molar entropy rows were then computed using:

$$=(1 - x)*hf + x*hg$$

with appropriate cell references substituted for each stream.

This systematic structure ensured that both ideal and actual state properties were calculated consistently.

3.3 Filling in the blanks with known data

The known pressures and temperatures of the streams were input into the respective cells. We know that stream 5 would contain the same temperature as stream 6, because the condenser does not change the temperature(significantly) of the stream, but only transforms its phase(from a liquid-vapor mixture to liquid)

Because SPL-100 is a simple splitter, all intensive properties are conserved:

$$(P_3, T_3, h_3, s_3) = (P_4, T_4, h_4, s_4) = (P_2, T_2, h_2, s_2)$$

Mass conservation gives:

$$\dot{n}_2 = \dot{n}_3 + \dot{n}_4$$

Defining the bleed fraction as

$$y = \frac{\dot{n}_3}{\dot{n}_2}$$

The molar flow rate relations were implemented in Excel using direct cell references. A single cell was designated for the bleed fraction y , and the outlet flow rates from the splitter were defined as:

$$\dot{n}_3 = y\dot{n}_2, \quad \dot{n}_4 = (1 - y)\dot{n}_2$$

This structure ensures automatic mass conservation and consistent propagation of flow changes throughout the cycle model. These functions will be implemented into excel, but will not have use until the final setup of the excel sheet. For now, the molar flow rate of stream 1 is set to an initial guess of 1 kmol/s The condenser outlet (Stream 6) is specified as saturated liquid at 40°C, giving

$$x_6 = 0, \quad P_6 = P_{sat}(40^\circ\text{C})$$

The saturation pressure was computed using the CHEcalculator function `=WaterPsat(T)`. Assuming negligible pressure drop across the condenser,

$$P_5 = P_6$$

The open feedwater heater (OFWH) is modeled as an adiabatic mixing chamber with no shaft work and negligible pressure losses. For steady-state mixing, mechanical equilibrium requires that all inlet and outlet streams be at the same pressure:

$$P_3 = P_7 = P_8$$

Since both inlet streams enter at 2 bar, the outlet stream must also be at 2 bar:

$$P_8 = 2 \text{ bar}$$

No pressure-changing device is present within the OFWH; therefore, the pressure remains constant across the unit. From question 2, we know that we want the outlet stream of the OFWH to be 1 degree celcius subcooled liquid, so the WaterTsat(P) function subtracted by 1, can provide the correct temperature.

In the CHEcalculator, with the streams where both the temperature and pressure are known, we can figure out if the fluid is a liquid, vapor, or liquid vapor mixture. The "Steam Properties" tool is used, and the known temperatures and pressures of a specific stream are inputted to provide either the liquid or vapor properties of the steam. With this we know that S1 is a vapor(x=1), and S8 is a liquid(x=0).

Streams 7 and 9 correspond to the outlet streams of pumps P-100 and P-101, respectively. Since pumps operate on liquid and do not induce phase change, the working fluid remains in the liquid phase during compression.

Stream 6 enters Pump P-100 as saturated liquid at 40°C. The pump increases the pressure to 2 bar without vaporization, resulting in a compressed (subcooled) liquid at Stream 7. Similarly, Pump P-101 raises the pressure from 2 bar to 40 bar, producing another compressed liquid at Stream 9.

Thus, the vapor fractions are set to

$$x_7 = 0$$

$$x_9 = 0$$

3.4 Extensive Property Rows: Enthalpy Flow, Entropy Flow, and Availability

After each stream's intensive thermodynamic properties (molar enthalpy and molar entropy) were calculated, the spreadsheet was extended by computing the corresponding *flow* quantities.

3.4.1 Enthalpy Flow

For each stream i , the enthalpy flow rate is defined as the product of the molar flow rate and the molar enthalpy:

$$\dot{H}_i = \dot{n}_i h_i$$

where $\dot{n}_i$ is the molar flow rate (kmol/s) and h_i is the molar enthalpy (kJ/kmol). The resulting units are kJ/s, which are equivalent to kW.

Excel implementation. In the *Enthalpy Flow (kJ/s)* row, each stream cell was calculated by multiplying the stream's molar flow cell by its molar enthalpy cell:

$$\text{Enthalpy Flow cell} = (\text{Molar Flow cell}) \times (\text{Molar Enthalpy cell})$$

3.4.2 Entropy Flow

For each stream i , the entropy flow rate is defined similarly:

$$\dot{S}_i = \dot{n}_i s_i$$

where s_i is the molar entropy (kJ/(kmol·K)). The resulting units are kJ/(s·K).

Excel implementation. In the *Entropy Flow (kJ/(s K))* row, each stream cell was calculated by multiplying the stream's molar flow cell by its molar entropy cell:

$$\text{Entropy Flow cell} = (\text{Molar Flow cell}) \times (\text{Molar Entropy cell})$$

3.4.3 Availability Flow

To quantify irreversibilities and lost work, the availability (flow exergy) rate for each stream i was computed relative to the environment (dead state). The specific flow exergy (per kmol) is given by:

$$\dot{B}_i = \dot{H}_i - T_0 \dot{S}_i$$

where T_0 is the temperature of the infinite environment, and h_0 and s_0 are the dead-state molar enthalpy and entropy, respectively, evaluated at the chosen reference state.

Excel implementation.

In the *Availability (kJ/s)* row, each stream cell was calculated using:

$$\text{Availability cell} = (\text{Enthalpy Flow cell}) - T_0 \times (\text{Entropy Flow cell})$$

With $T_0 = 300$ K placed in another cell, to be referenced by the equation above.

Stream Properties	S1	S2IDEAL	S2	S3	S4	SSIDEAL	S5	S6	STIDEAL	S7	S8	SSIDEAL	S9	S1
Molar Flow, kmol/s	1.0000	1.0000	1.0000	0.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Mass Flow, kg/s														
Volume Flow, m ³ /s														
Temperature, C	450.0000			0.0000	0.0000	40.0000	40.0000	40.0000			119.2308			450.0000
Pressure, bar	40.0000	2.0000	2.0000	2.0000	2.0000	0.0738	0.0738	0.0738	2.0000	2.0000	2.0000	40.0000	40.0000	40.0000
Vapor Fraction								0.0000		0.0000	0.0000	0.0000	0.0000	
Molecular Weight, kg/kmol														
Molar Enthalpy, kJ/kmol	60011.902	9092.186	9092.186	9092.186	9092.186	46377.193	46377.193	46377.193	9092.186	9092.186	9015.728	19589.546	19589.546	60011.902
Vapor Phase	60011.902	48753.821	48753.821	48753.821	48753.821	46377.193	46377.193	46377.193	48753.821	48753.821	9015.728	50448.097	50448.097	60011.902
Liquid Phase	60011.902	9092.186	9092.186	9092.186	9092.186	46377.193	46377.193	46377.193	9092.186	9092.186	9015.728	19589.546	19589.546	60011.902
Molar Entropy, kJ/(kmol K)	125.003	27.564	27.564	27.564	27.564	148.773	148.773	148.773	27.564	27.564	27.370	50.379	50.379	125.003
Vapor Phase	125.003	128.390	128.390	128.390	128.390	148.773	148.773	148.773	128.390	128.390	27.370	109.324	109.324	125.003
Liquid Phase	125.003	27.564	27.564	27.564	27.564	148.773	148.773	148.773	27.564	27.564	27.370	50.379	50.379	125.003
Molar Density, kmol/m ³														
Enthalpy Flow, kJ/s	60011.90	9092.19	9092.19	0.00	9092.19	46377.19	46377.19	46377.19	9092.19	9092.19	9015.73	19589.55	19589.55	60011.90
Entropy Flow, kJ/(s K)	125.00	27.56	27.56	0.00	27.56	148.77	148.77	148.77	27.56	27.56	27.37	50.38	50.38	125.00
Availability, kJ/s	22511.07	822.85	822.85	0.00	822.85	1745.44	1745.44	1745.44	822.85	822.85	804.77	4475.73	4475.73	22511.07
Duty [Power], kJ/s [kW]														
Efficiency														

Figure 1: State of the Excel Sheet after Setup and Property Calculations

4 Unit Operation Analysis

4.1 Turbine Analysis

The turbine is modeled as a steady-flow control volume. The general energy balance for a control volume is:

$$\frac{dE_{CV}}{dt} = \sum \dot{Q} - \sum \dot{W} + \sum \dot{m}_{in} \left(h + \frac{V^2}{2} + gz \right) - \sum \dot{m}_{out} \left(h + \frac{V^2}{2} + gz \right)$$

For steady-state operation,

$$\frac{dE_{CV}}{dt} = 0$$

Assuming negligible kinetic and potential energy changes and minimal heat losses (adiabatic operation), the energy balance simplifies to:

$$0 = \dot{W}_t + \sum \dot{m}_{in} h_{in} - \sum \dot{m}_{out} h_{out}$$

Rearranging,

$$\dot{W}_t = \sum \dot{m}_{out} h_{out} - \sum \dot{m}_{in} h_{in}$$

For a single-inlet, single-outlet turbine, this reduces to:

$$\dot{W}_t = \dot{m}(h_{out} - h_{in})$$

For the convenience of the information given and the desired results, we will convert this equation into the molar form:

$$\dot{W}_t = \dot{n}(h_{out} - h_{in})$$

Because the enthalpy flow rate for each stream is already defined in the spreadsheet as

$$\dot{H}_i = \dot{n}_i h_i$$

the turbine power can be written directly in terms of enthalpy flow:

$$\dot{W}_t = \dot{H}_{out} - \dot{H}_{in}$$

Excel implementation.

In the spreadsheet, the turbine duty (power) cell is computed directly as:

$$= (\text{EnthalpyFlow_outlet_cell}) - (\text{EnthalpyFlow_inlet_cell})$$

These equations are implemented under the S2 and S5 columns, along with their ideal counterparts, which will be needed to calculate the turbine efficiency.

The turbine efficiency provided in the problem statement ($\eta_t = 0.85$) is interpreted as the *isentropic efficiency* of an adiabatic turbine. It is defined as the ratio of the actual turbine work output to the ideal (reversible, adiabatic) work output for the same inlet state and outlet pressure:

$$\eta_t = \frac{w_{act}}{w_{ideal}}$$

For now the cell for efficiency for the HP and LP turbines will contain the previous equation by referencing the actual and ideal work, but right the values have no meaning yet, as there is no actual ideal process yet set.

The ideal turbine process is reversible and adiabatic, and therefore isentropic:

$$s_{out,s} = s_{in}$$

Now that we have this entropy balance, a cell in this sheet will contain the following equation will be implemented into corresponding cells - this will be our entropy balance for both the HP and LW turbines

$$= s_{out,s} - s_{in}$$

We will first start off with the HP turbine(Streams 1-2). We know that the value within the cell for the entropy balance has to be equivalent to 0, and we know that either the temperature or vapor fraction have an impact on the entropy value that we calculate. We will start off with a guess for the vapor fraction(between 0-1), and then implement excel's built in goal seek function. We will set the cell containing the entropy balance to 0, by changing the vapor fraction of Stream 2 ideal.

The turbine isentropic efficiency is given as

$$\eta_t = 0.85$$

The turbine efficiency is defined as

$$\eta_t = \frac{w_{actual}}{w_{ideal}} = \frac{h_{in} - h_{out}}{h_{in} - h_{out,s}}$$

To implement this relation in Excel, a residual equation was constructed:

$$\frac{w_{actual}}{w_{ideal}} - 0.85 = 0$$

Goal Seek was then used to set this residual equal to zero by varying the vapor fraction of the turbine outlet stream.

When Goal Seek was initially performed under the assumption of a two-phase outlet state, the resulting vapor fraction was found to be greater than one which indicates that the outlet state does not lie in the two-phase region. Instead, the fluid must be in the superheated vapor region.

Accordingly, the vapor fraction was fixed at

$$x = 1,$$

and the outlet state was redefined as superheated vapor at the specified outlet pressure.

In the superheated region, temperature becomes the appropriate independent variable. A reasonable initial guess for the outlet temperature was entered into the corresponding temperature cell for the turbine outlet stream.

Goal Seek was then executed again by setting the turbine efficiency residual

$$\frac{w_{actual}}{w_{ideal}} - 0.85$$

equal to zero, this time varying the outlet temperature.

Now for the LP turbine, the pressure dependencies for the enthalpy and entropy values were removed, since the stream would be a liquid-vapor mixture. The same goal seeks as the HP turbine were utilized, and the results for the vapor fractions were both in the range $0 < x < 1$, which means that no alterations are needed.

Figure 2 and Figure 3 display the results given by the following goal seek calculations of the HP and LP turbines. The cells highlighted in yellow are cells in which we input our initial guess, and to be adjusted by the goal seek function (in the figures, the goal seek operations were already performed, except for the molar flow of stream 1).

Stream Properties	S1	S2IDEAL	S2
Molar Flow, kmol/s	1.0000	1.0000	1.0000
Mass Flow, kg/s			
Volume Flow, m ³ /s			
Temperature, C	450.0000		134.8993
Pressure, bar	40.0000	2.0000	2.0000
Vapor Fraction	1.0000	0.9664	1.0000
Molecular Weight, kg/kmol			
Molar Enthalpy, kJ/kmol	60011.902	47421.451	49310.016
Vapor Phase	60011.902	48753.821	49310.016
Liquid Phase	60011.902	9092.186	49310.016
Molar Entropy, kJ/(kmol K)	125.003	125.003	129.778
Vapor Phase	125.003	128.390	129.778
Liquid Phase	125.003	27.564	129.778
Molar Density, kmol/m ³			
Enthalpy Flow, kJ/s	60011.90	47421.45	49310.02
Entropy Flow, kJ/(s K)	125.00	125.00	129.78
Availability, kJ/s	22511.07	9920.62	10376.59
Duty [Power], kJ/s [kW]		-12590.45	-10701.89
Efficiency			0.85

Figure 2: HP turbine results

Stream Properties	S4	SSIDEAL	S5
Molar Flow, kmol/s	1.0000	1.0000	1.0000
Mass Flow, kg/s			
Volume Flow, m ³ /s			
Temperature, C	134.8993	40.0000	40.0000
Pressure, bar	2.0000	0.0738	0.0738
Vapor Fraction	1.0000	0.8628	0.8935
Molecular Weight, kg/kmol			
Molar Enthalpy, kJ/kmol	49310.016	40429.083	41761.223
Vapor Phase	49310.016	46377.193	46377.193
Liquid Phase	49310.016	3016.658	3016.658
Molar Entropy, kJ/(kmol K)	129.778	129.778	134.032
Vapor Phase	129.778	148.773	148.773
Liquid Phase	129.778	10.307	10.307
Molar Density, kmol/m ³			
Enthalpy Flow, kJ/s	49310.02	40429.08	41761.22
Entropy Flow, kJ/(s K)	129.78	129.78	134.03
Availability, kJ/s	10376.59	1495.66	1551.60
Duty [Power], kJ/s [kW]		-8880.93	-7548.79
Efficiency			0.85

Figure 3: LP turbine results

4.2 Feedwater Pump Analysis

Energy Balance

The feedwater pump is modeled as a steady-flow control volume. The general energy balance is:

$$\frac{dE_{CV}}{dt} = \sum \dot{Q} - \sum \dot{W} + \sum \dot{m}_{in} \left(h + \frac{V^2}{2} + gz \right) - \sum \dot{m}_{out} \left(h + \frac{V^2}{2} + gz \right)$$

For steady-state operation,

$$\frac{dE_{CV}}{dt} = 0$$

Assuming negligible kinetic and potential energy changes and minimal heat transfer (adiabatic operation), the balance simplifies to:

$$0 = -\dot{W}_p + \dot{m}(h_{in} - h_{out})$$

Rearranging,

$$\dot{W}_p = \dot{m}(h_{out} - h_{in})$$

Since the spreadsheet model uses molar flow rates, this becomes:

$$\dot{W}_p = \dot{n}(h_{out} - h_{in})$$

Because enthalpy flow rates are already defined as

$$\dot{H}_i = \dot{n}_i h_i,$$

the pump power can be computed directly as:

$$\dot{W}_p = \dot{H}_{out} - \dot{H}_{in}$$

Excel implementation.

In the spreadsheet, the pump power input is calculated as:

$$= (\text{EnthalpyFlow_outlet_cell}) - (\text{EnthalpyFlow_inlet_cell})$$

As previously established in the entropy balance for the turbines, the same formulation can be applied to the feedwater pumps. As before, under steady-state and adiabatic conditions, the entropy balance reduces to:

$$s_{out,s} = s_{in}$$

For the ideal pump, the process is assumed reversible and adiabatic; therefore, entropy generation is zero and the process is isentropic.

A cell in the spreadsheet is used to enforce the entropy balance for the ideal pump stream. The implemented relation is:

$$s_{out,s} - s_{in}$$

This residual cell is set equal to zero using Excel's Goal Seek function by varying the outlet temperature of the ideal stream at the specified outlet pressure.

Because Streams 7 and 9 correspond to compressed liquid states, vapor quality is not a valid independent variable. At fixed pressure, temperature is therefore the only adjustable thermodynamic variable that directly affects the molar enthalpy and entropy.

Because Stream 6 is a saturated liquid, its thermodynamic properties depend on a single independent variable due to the saturation relation $P = P_{sat}(T)$. Thus, enthalpy and entropy are evaluated directly as saturated liquid properties at the specified temperature, and those changes are implemented into the spreadsheet.

After implementing these modifications, the Goal Seek function was executed to enforce the isentropic condition. The entropy residual,

$$s_{out,s} - s_{in},$$

was set equal to zero by varying the outlet temperature of the ideal Stream 7 at the specified pressure.

The isentropic efficiency of a pump differs from that of a turbine because a pump requires work input rather than producing work output.

The isentropic efficiency of the pump is defined as the ratio of the ideal work input to the actual work input:

$$\eta_p = \frac{w_{ideal}}{w_{actual}}$$

This relation is implemented in the spreadsheet to determine the efficiency of each pump.

A residual cell was constructed as

$$\eta_{actual} - 0.75,$$

where 0.75 is the specified pump efficiency. Goal Seek was then used to drive this residual to zero by varying the outlet temperature of the actual stream at the specified pressure.

The same methodology was applied to the pump between Streams 8 and 9, enforcing the isentropic condition for the ideal state and using Goal Seek to satisfy the specified pump efficiency. In figures 4 and 5, the results from the previously explained instructions for both pumps in the cycle are displayed. In the yellow highlighted cells are the cells that were changed by goalseek.

Stream Properties	S6	S7IDEAL	S7
Molar Flow, kmol/s	1.0000	1.0000	1.0000
Mass Flow, kg/s			
Volume Flow, m ³ /s			
Temperature, C	40.0000	40.0057	40.0211
Pressure, bar	0.0738	2.0000	2.0000
Vapor Fraction	0.0000	0.0000	0.0000
Molecular Weight, kg/kmol			
Molar Enthalpy, kJ/kmol	3016.658	3020.155	3021.321
Vapor Phase			
Liquid Phase	3016.658	3020.155	3021.321
Molar Entropy, kJ/(kmol K)	10.307	10.307	10.311
Vapor Phase			
Liquid Phase	10.307	10.307	10.311
Molar Density, kmol/m³			
Enthalpy Flow, kJ/s	3016.66	3020.15	3021.32
Entropy Flow, kJ/(s K)	10.31	10.31	10.31
Availability, kJ/s	-75.38	-71.89	-71.84
Duty [Power], kJ/s [kW]		-3.50	-4.66
Efficiency			0.75

Figure 4: Condensate Pump results

Stream Properties	S8	S9IDEAL	S9
Molar Flow, kmol/s	1.0000	1.0000	1.0000
Mass Flow, kg/s			
Volume Flow, m ³ /s			
Temperature, C	119.2308	119.5480	119.8648
Pressure, bar	2.0000	40.0000	40.0000
Vapor Fraction	0.0000	0.0000	0.0000
Molecular Weight, kg/kmol			
Molar Enthalpy, kJ/kmol	9015.728	9088.222	9112.387
Vapor Phase			
Liquid Phase	9015.728	9088.222	9112.387
Molar Entropy, kJ/(kmol K)	27.370	27.370	27.431
Vapor Phase			
Liquid Phase	27.370	27.370	27.431
Molar Density, kmol/m³			
Enthalpy Flow, kJ/s	9015.73	9088.22	9112.39
Entropy Flow, kJ/(s K)	27.37	27.37	27.43
Availability, kJ/s	804.77	877.27	882.98
Duty [Power], kJ/s [kW]		72.49	96.66
Efficiency			0.75

Figure 5: Feed Pump results

4.3 Boiler Energy Balance

The boiler is modeled as a steady-flow control volume. The general energy balance for a control volume is:

$$\frac{dE_{CV}}{dt} = \sum \dot{Q} - \sum \dot{W} + \sum \dot{m}_{in} \left(h + \frac{V^2}{2} + gz \right) - \sum \dot{m}_{out} \left(h + \frac{V^2}{2} + gz \right)$$

For steady-state operation,

$$\frac{dE_{CV}}{dt} = 0$$

In the boiler, no shaft work is performed:

$$\dot{W} = 0$$

Additionally, kinetic and potential energy effects are assumed negligible. Under these assumptions, the energy balance reduces to:

$$\dot{Q}_{in} = \sum \dot{m}_{out} h_{out} - \sum \dot{m}_{in} h_{in}$$

For a single-inlet, single-outlet boiler, this simplifies to:

$$\dot{Q}_{in} = \dot{m}(h_{out} - h_{in})$$

Since the spreadsheet is formulated in terms of molar flow rates,

$$\dot{Q}_{in} = \dot{n}(h_{out} - h_{in})$$

Because enthalpy flow rates are already defined as

$$\dot{H}_i = \dot{n}_i h_i,$$

the boiler heat input can be calculated directly as:

$$\dot{Q}_{in} = \dot{H}_{out} - \dot{H}_{in}$$

4.4 Open Feedwater Heater (OFWH) Mixing Model (Q2)

The open feedwater heater (OFWH) is modeled as a steady-state, adiabatic mixing chamber with no shaft work and negligible kinetic and potential energy effects. Stream 3 (bled steam) mixes directly with Stream 7 (condensate after Pump P-100) to form Stream 8.

Molar Balance

Define the bleed fraction y as the fraction of the HP turbine discharge (Stream 2) extracted into Stream 3:

$$\dot{n}_3 = y \dot{n}_2$$

Because SPL-100 splits Stream 2 into Streams 3 and 4:

$$\dot{n}_4 = (1 - y)\dot{n}_2$$

Since Stream 4 expands through the LP turbine and then condenses and is pumped to Stream 7 (no split/merge in between), the molar flow remains the same:

$$\dot{n}_7 = \dot{n}_4 = (1 - y)\dot{n}_2$$

The OFWH mixing balance gives:

$$\dot{n}_3 + \dot{n}_7 = \dot{n}_8$$

Substituting the relations above:

$$y\dot{n}_2 + (1 - y)\dot{n}_2 = \dot{n}_8 \quad \Rightarrow \quad \dot{n}_8 = \dot{n}_2$$

Energy Balance (Adiabatic Mixing)

For the OFWH at steady state with $\dot{Q} \approx 0$ and $\dot{W} = 0$:

$$\sum \dot{n}_{in} h_{in} = \sum \dot{n}_{out} h_{out}$$

Thus:

$$\dot{n}_3 h_3 + \dot{n}_7 h_7 = \dot{n}_8 h_8$$

Substitute $\dot{n}_3 = y\dot{n}_2$, $\dot{n}_7 = (1 - y)\dot{n}_2$, and $\dot{n}_8 = \dot{n}_2$:

$$y\dot{n}_2 h_3 + (1 - y)\dot{n}_2 h_7 = \dot{n}_2 h_8$$

Solve for y :

$$\boxed{y = \frac{h_8 - h_7}{h_3 - h_7}}$$

This expression was used to determine the bleed fraction required to achieve the specified OFWH outlet condition (Stream 8). The bleed fraction cell for Stream 3 was defined using the derived enthalpy relation with appropriate cell references. Since all flow rates were expressed as functions of y , updates to this value automatically adjusted the downstream stream flow rates.

Substitution of the calculated enthalpy values into the OFWH balance equation and implementation in Excel yields

$$y = 0.1295$$

Thus, approximately 12.95% of the HP turbine discharge must be extracted to satisfy the required feedwater heater outlet condition. This completes the solution to Question 2.

5 Determination of Steam Flow Rate (Q1)

The only components producing shaft work in the cycle are the HP and LP turbines. Therefore, the net turbine power output is given by:

$$\dot{W}_{t,net} = \dot{W}_{HP} + \dot{W}_{LP}$$

To satisfy the required power output, the following residual equation was constructed in the spreadsheet:

$$\dot{W}_{HP} + \dot{W}_{LP} - 500 = 0$$

The goal is to determine the steam flow rate that satisfies this condition.

Because turbine work is proportional to molar flow rate,

$$\dot{W} = \dot{n}(h_{in} - h_{out}),$$

adjusting the inlet molar flow rate directly scales the power output of both turbines.

Accordingly, Excel's Goal Seek function was used to set the power residual equal to zero by varying the molar flow rate of Stream 1.

Since all downstream stream flow rates are defined as functions of $\dot{n}_1$ (including the bleed fraction relation),

$$\dot{n}_3 = y\dot{n}_2,$$

any change in the inlet molar flow automatically propagates throughout the entire cycle model. This ensures that the resulting value of $\dot{n}_1$ corresponds to the steam flow rate required to produce exactly 500 kW of turbine power.

After solving for the required inlet molar flow rate in kmol/s, the value was converted to MMSCFD using the calculator's built-in conversion function:

$$=\text{KMOLperSECToMMSCFD}(\text{Stream1_MolarFlow_Cell})$$

This function references the molar flow rate of Stream 1 and converts the value to million standard cubic feet per day (MMSCFD).

The resulting steam flow rate required to produce 500 kW of power is:

$$\dot{V}_{steam} = 2.0923 \text{ MMSCFD}$$

6 Thermal Efficiency of the Cycle (Q3)

The thermal efficiency of the Rankine cycle is defined as the ratio of net work output to heat input:

$$\eta_{th} = \frac{\dot{W}_{net}}{\dot{Q}_{in}}$$

In this system, shaft work is produced by the HP and LP turbines. Therefore,

$$\dot{W}_{net} = \dot{W}_{HP} + \dot{W}_{LP}$$

The heat input to the working fluid is provided by the boiler:

$$\dot{Q}_{in} = \dot{H}_{out,boiler} - \dot{H}_{in,boiler}$$

However, the boiler efficiency is specified as 85%. This means that only 85% of the fuel energy supplied to the boiler is transferred to the steam. Thus, the required fuel energy input is:

$$\dot{Q}_{fuel} = \frac{\dot{Q}_{in}}{0.85}$$

Accordingly, the thermal efficiency relative to the fuel input becomes:

$$\eta_{th} = \frac{\dot{W}_{HP} + \dot{W}_{LP}}{\dot{Q}_{fuel}} = \frac{\dot{W}_{HP} + \dot{W}_{LP}}{\frac{\dot{Q}_{in}}{0.85}}$$

Excel implementation.

In the spreadsheet, the numerator is computed as the sum of the HP and LP turbine power outputs, and the denominator is calculated as the boiler heat input divided by 0.85. The thermal efficiency cell therefore contains:

$$= \text{ABS}(W_HP + W_LP) / (Q_in / 0.85)$$

After implementing the efficiency expression in Excel, the result was formatted using Excel's percentage function to present the value in percent form.

The calculated thermal efficiency of the cycle is:

$$\boxed{\eta_{th} = 29\%}$$

7 Lost Work Formulation(Q4)

The lost work for a control volume can be written in general form:

$$LW = -\Delta \dot{m}(H - T_0 S) + \sum \dot{Q}_i \left(1 - \frac{T_0}{T_i}\right) + \sum \dot{W}_s$$

where T_0 is the dead-state temperature, $\dot{Q}_i$ is heat transfer crossing the boundary at temperature T_i , and $\dot{W}_s$ represents shaft work interactions.

Because the term $(H - T_0 S)$ represents flow availability, the availability rate can be defined as:

$$\dot{B} = \dot{H} - T_0 \dot{S}$$

Thus, the net change in availability across the control volume can be expressed using availability in and out:

$$\Delta \dot{m} (H - T_0 S) = \dot{B}_i - \dot{B}_e$$

where:

- $\dot{B}_i$ = total availability rate entering the control volume
- $\dot{B}_e$ = total availability rate exiting the control volume

Substituting this into the lost work equation yields the equivalent form:

$$LW = (\dot{B}_i - \dot{B}_e) + \sum_j \dot{Q}_j \left(1 - \frac{T_0}{T_i}\right) + \sum_k \dot{W}_s$$

7.1 Lost Work Expressions for Each Unit Operation

Starting from the general lost work relation written in availability form,

$$LW = (\dot{B}_i - \dot{B}_e) + \sum_i \dot{Q}_i \left(1 - \frac{T_0}{T_i}\right) + \sum_s \dot{W}_s$$

the appropriate terms can be removed or retained depending on the operating assumptions for each unit operation.

Turbine (Adiabatic)

For the turbines, heat transfer is assumed negligible, so the heat term is removed. The lost work becomes:

$$LW_{\text{turb}} = (\dot{B}_{in} - \dot{B}_{out}) + \dot{W}_t$$

Condenser

For the condenser, there is no shaft work, and heat transfer occurs to the cooling water at a boundary temperature T_i . Thus:

$$LW_{\text{cond}} = (\dot{B}_{in} - \dot{B}_{out}) + \dot{Q}_{out} \left(1 - \frac{T_0}{T_c}\right)$$

Pump

For pumps, heat transfer is assumed negligible, but shaft work is required as input. Therefore:

$$LW_{\text{pump}} = (\dot{B}_{in} - \dot{B}_{out}) + \dot{W}_p$$

Open Feedwater Heater (OFWH)

The OFWH is modeled as an adiabatic mixing chamber with no shaft work. Therefore, both the heat and shaft-work terms are zero, leaving only the availability difference. Because there are two inlet streams and one outlet stream:

$$LW_{\text{OFWH}} = (\dot{B}_i - \dot{B}_e)$$

with

$$\dot{B}_i = \dot{B}_3 + \dot{B}_7, \quad \dot{B}_e = \dot{B}_8,$$

so:

$$LW_{\text{OFWH}} = (\dot{B}_3 + \dot{B}_7) - \dot{B}_8$$

Boiler

For the boiler, there is no shaft work, but heat transfer occurs from the combustion products to the working fluid at a boundary temperature T_i . Thus:

$$LW_{\text{boiler}} = (\dot{B}_{in} - \dot{B}_{out}) + \dot{Q}_{in} \left(1 - \frac{T_0}{T_b}\right)$$

7.2 Spreadsheet Implementation of Lost Work Relations

The lost work expressions derived for each unit operation were implemented directly in Excel using cell-referenced formulas. For each component, the appropriate availability terms ($\dot{B}$), heat-transfer terms, and shaft-work terms were computed by referencing the corresponding stream and duty cells in the previously established worksheet. This approach ensured that any updates to stream properties or flow rates automatically propagated through the lost work calculations.

Boiler Heat Input Adjustment

For the boiler, the heat-transfer exergy term was evaluated using the specified heat-source temperature of $T_i = 1250$ K. In addition, the boiler efficiency is given as 85%, meaning that the required fuel energy input exceeds the heat transferred to the working fluid. Therefore, the effective heat input used in the lost work relation was adjusted as:

$$\dot{Q}_{fuel} = \frac{\dot{Q}_{in}}{0.85}$$

Accordingly, the boiler lost work expression was implemented in Excel in the form:

$$LW_{\text{boiler}} = (\dot{B}_{in} - \dot{B}_{out}) + \left(\frac{\dot{Q}_{in}}{0.85}\right) \left(1 - \frac{T_0}{1250}\right)$$

where T_0 is the dead-state temperature used throughout the analysis.

After being put into excel, the lost work, in kW, for each unit was displayed as seen in Figure 6, and highlighted in yellow is the total lost work. Lost work is important to every process as it represents the

exergy destroyed due to irreversibilities and serves as a direct indicator of how far a real process deviates from ideal reversible performance.

Steam Flow, MMSCFD	2.0923
Turbine Lost Work(1-2), kW	41.4689
Turbine Lost Work(4-5), kW	32.1578
Condenser Lost Work(5-6), kW	40.9969
Pump Lost Work(6-7), kW	0.0281
OFWH Lost Work (3,7-8), kW	13.7921
Pump Lost Work(8-9), kW	0.5342
Boiler Lost Work(9-1), kW	691.3080
Lost Work Total, kW	820.2861

Figure 6: Table with the list of all equipment and the lost work associated with each one, Steam Flow in yellow is unrelated

8 Thermodynamic Efficiency of The Power Plant (Q5)

The thermodynamic efficiency of the plant is defined as the ratio of useful work output to the total available energy input. In terms of lost work, this may be expressed as:

$$\eta_{II} = \frac{\dot{W}_{useful}}{\dot{W}_{useful} + LW_{total}}$$

In this system, the useful work output is produced by the two turbines:

$$\dot{W}_{useful} = \dot{W}_{HP} + \dot{W}_{LP}$$

Thus, the thermodynamic efficiency becomes:

$$\eta_{II} = \frac{\dot{W}_{HP} + \dot{W}_{LP}}{\dot{W}_{HP} + \dot{W}_{LP} + LW_{total}}$$

Excel implementation.

In the spreadsheet, the numerator was computed as the sum of the HP and LP turbine power outputs. The denominator was calculated as this same total turbine power plus the total lost work previously determined for the entire cycle.

The resulting value was then formatted using Excel's percentage function to present the final thermodynamic efficiency in percent form.

Evaluating the expression in Excel and formatting the result using the percentage function yields:

$$\boxed{\eta_{II} = 38\%}$$

Thus, the thermodynamic efficiency of the plant is 38%.

9 Maximum Theoretical Efficiency(Q6)

The maximum theoretical efficiency for any heat engine operating between a heat source at temperature T_H and a heat sink at temperature T_0 is given by the Carnot efficiency:

$$\eta_{\max} = 1 - \frac{T_0}{T_H}$$

For this plant, the heat sink (cooling water/environment) temperature is $T_0 = 300$ K and the heat source temperature is $T_H = 1250$ K. Substituting:

$$\eta_{\max} = 1 - \frac{300}{1250} = 0.76$$

Thus, the maximum theoretical efficiency is:

$$\boxed{\eta_{\max} = 76\%}$$

10 Annual Operating Cost (Q7)

10.1 Annual Operating Cost – Electricity

Electricity must be purchased to operate the feedwater pumps. The annual electricity cost is calculated as the product of pump power consumption, the unit cost of electricity, and the annual operating hours.

The cost expression is:

$$\text{Cost}_{elec} = (\dot{W}_{P1} + \dot{W}_{P2}) \times \left(\frac{\$}{\text{kWh}} \right) \times (\text{operating hours per year})$$

Since pump power is expressed in kW, multiplying by operating hours yields total energy consumption in kWh per year. This value is then multiplied by the electricity rate to determine the annual cost.

Evaluating this expression in Excel gives:

\$1,807.59 per operating year

10.2 Annual Operating Cost – Cooling Water

Cooling water is used in the condenser to remove heat from the cycle. For the condenser, there is no shaft work, and kinetic and potential energy effects are neglected. Under steady-state conditions, the energy balance reduces to:

$$\dot{Q}_{out} = \dot{H}_{in} - \dot{H}_{out}$$

where $\dot{Q}_{out}$ is the rate of heat rejected by the condenser (kW).

To compute the annual cooling water utility cost, the condenser heat rejection rate is converted from kW (kJ/s) to GJ/s and multiplied by the cooling water price (\$/GJ). The result is then multiplied by the annual operating time in seconds:

$$t_{op} = (8000 \text{ hr/yr}) \times (60 \text{ min/hr}) \times (60 \text{ s/min})$$

Thus, the annual cooling water cost is calculated as:

$$\text{Cost}_{cw} = \left(\dot{Q}_{out} \times \frac{1 \text{ GJ}}{10^6 \text{ kJ}} \right) \left(\frac{\$}{\text{GJ}} \right) t_{op}$$

Evaluating this expression in Excel yields an annual cooling water cost of:

\$9,953.47 per operating year

10.3 Annual Operating Cost – Natural Gas

Natural gas is used to supply heat to the boiler. The boiler heat transfer rate to the working fluid is obtained from the boiler energy balance as $\dot{Q}_{in}$ (kW). To determine the corresponding fuel energy required, the boiler efficiency must be included. With $\eta_{boiler} = 0.85$,

$$\dot{Q}_{fuel} = \frac{\dot{Q}_{in}}{0.85}$$

The annual natural gas cost is then calculated by converting the fuel heat rate from kW (kJ/s) to GJ/s and multiplying by the natural gas unit price (\$/GJ) and the annual operating time in seconds:

$$t_{op} = (8000 \text{ hr/yr}) \times (60 \text{ min/hr}) \times (60 \text{ s/min})$$

Thus,

$$\text{Cost}_{NG} = \left(\frac{\dot{Q}_{in}}{0.85} \right) \left(\frac{1 \text{ GJ}}{10^6 \text{ kJ}} \right) \left(\frac{\$}{\text{GJ}} \right) t_{op}$$

Evaluating this expression in Excel yields an annual natural gas cost of:

\$212,166.00 per operating year

10.4 Total Annual Operating Cost

The total operating cost per year was calculated by summing the annual costs for electricity, cooling water, and natural gas. In Excel, each of these cost cells was formulated using cell references to previously computed duties, conversion factors, unit costs, and operating time, ensuring that no hard-coded numerical values were embedded directly within the cost equations. This fully referenced structure allows the total cost to update automatically if any upstream assumptions or calculated values change.

The resulting total cost per operating year is:

\$223,927.07 per operating year

10.5 Cost Optimization via HP Turbine Discharge Pressure (Q8)

The objective is to minimize the annual operating cost by varying the discharge pressure of the HP turbine. Evaluating many candidate pressures manually would be time-intensive, so the process was automated within Excel.

A key requirement for automation is that the property calculations remain valid regardless of whether a stream lies in the two-phase region or in the superheated region. To accomplish this, the molar enthalpy and molar entropy cells for Streams 1–4 were written using conditional logic that switches between the lever rule (two-phase) and the superheated property functions.

Conditional Enthalpy Calculation (Streams 1–4)

For each stream, the molar enthalpy cell was defined as:

=IF(B19>=1, HSteam(B17,B18), B23*B19 + (1-B19)*B24)

This expression operates as follows:

- If the vapor fraction satisfies $x \geq 1$, the stream is treated as superheated vapor and the enthalpy is evaluated using `HSteam(T,P)` with dependencies on temperature and pressure.
- If $0 \leq x < 1$, the stream is treated as a two-phase mixture and the lever rule is applied:

$$h = xh_g + (1 - x)h_f$$

using the saturated vapor and saturated liquid enthalpies referenced in the worksheet.

Because the cell references were structured consistently by stream, this conditional enthalpy equation was dragged across Streams 1–4.

Conditional Entropy Calculation (Streams 1–4)

The molar entropy cells were defined using the same structure, with the only change being that entropy is evaluated with the `=SSteam()` function in the superheated case. The lever rule form remains unchanged:

$$s = xs_g + (1 - x)s_f$$

Thus, the entropy equation was likewise copied across Streams 1–4 with only the function name and referenced entropy cells adjusted accordingly. A VBA macro was developed to automate the Goal Seek operations required to solve the system at different HP turbine discharge pressures. The code is presented in Figures 7 and 8.

Although the macro significantly reduces manual intervention in solving the thermodynamic balances, it currently requires the HP discharge pressure to be adjusted manually before execution. After updating the pressure in the worksheet, the macro is run, and the resulting total annual cost is recorded along with the corresponding pressure value.

The code could be further enhanced to automatically iterate over a defined pressure range and export the computed costs to a separate sheet for streamlined optimization and plotting.

```
(General)
Sub VBAcodeForProblem8Assignment2()
    'Main driver routine
    SolveRemainingUnits
End Sub

'Attempts solution assuming mixture-to-vapor case
Function MixToVapor() As Boolean
    On Error GoTo FailHandler

    'Initialize guesses
    Range("C17").Value = ""
    Range("C19").Value = 0.9
    Range("D17").Value = 300
    Range("D19").Value = 1

    'Solve first stream property (residual in B39)
    If Not Range("B39").GoalSeek(Goal:=0, ChangingCell:=Range("C19")) Then GoTo FailHandler

    'Solve second stream property (residual in B40)
    If Not Range("B40").GoalSeek(Goal:=0, ChangingCell:=Range("D17")) Then GoTo FailHandler

    MixToVapor = True
    Exit Function

FailHandler:
    MixToVapor = False
End Function

'Attempts solution assuming mixture-to-mixture case
Function MixToMix() As Boolean
    On Error GoTo FailHandler

    'Initialize guesses
    Range("C17").Value = ""
    Range("C19").Value = 0.9
    Range("D17").Value = ""
    Range("D19").Value = 0.9

    'Solve first stream property
    If Not Range("B39").GoalSeek(Goal:=0, ChangingCell:=Range("C19")) Then GoTo FailHandler

    'Solve second stream property
    If Not Range("B40").GoalSeek(Goal:=0, ChangingCell:=Range("D19")) Then GoTo FailHandler

    MixToMix = True
    Exit Function

FailHandler:
    MixToMix = False
End Function

'Attempts solution assuming vapor-to-vapor case
Function VaporToVapor() As Boolean
    On Error GoTo FailHandler

    'Initialize guesses
    Range("C17").Value = 300
    Range("C19").Value = 1
    Range("D17").Value = 300
    Range("D19").Value = 1
```

Figure 7: Part 1 of the VBA code used

```

'Solve first stream temperature
If Not Range("B39").GoalSeek(Goal:=0, ChangingCell:=Range("C17")) Then GoTo FailHandler

'Solve second stream temperature
If Not Range("B40").GoalSeek(Goal:=0, ChangingCell:=Range("D17")) Then GoTo FailHandler

VaporToVapor = True
Exit Function

FailHandler:
    VaporToVapor = False
End Function

```

```

'Solves remaining units after phase case is determined
Sub SolveRemainingUnits()

    'Select solution path based on C18 value
    If Range("C18").Value < 1.81 Then

        'Try mix-to-mix first
        If Not MixToMix() Then
            If Not MixToVapor() Then
                VaporToVapor
            End If
        End If

    ElseIf Range("C18").Value < 3.61 Then

        'Try mix-to-vapor first
        If Not MixToVapor() Then
            VaporToVapor
        End If

    Else

        'Default to vapor-to-vapor
        VaporToVapor

    End If

    'ST-101 ideal turbine (isentropic condition)
    Range("B42").GoalSeek Goal:=0, ChangingCell:=Range("G19")

    'ST-101 actual turbine (efficiency condition)
    Range("B43").GoalSeek Goal:=0, ChangingCell:=Range("H19")

    'P-100 ideal pump (isentropic condition)
    Range("B45").GoalSeek Goal:=0, ChangingCell:=Range("J17")

    'P-100 actual pump (efficiency condition)
    Range("B46").GoalSeek Goal:=0, ChangingCell:=Range("K17")

    'P-101 ideal pump
    Range("B47").GoalSeek Goal:=0, ChangingCell:=Range("M17")

    'Shift initial guess slightly for convergence
    Range("N17").Value = Range("M17").Value + 5

    'P-101 actual pump
    Range("B48").GoalSeek Goal:=0, ChangingCell:=Range("N17")

    'Final system balance (total output constraint)

```

```

Range("B50").GoalSeek Goal:=0, ChangingCell:=Range("B14")

End Sub

```

The annual operating cost was first evaluated at integer HP discharge pressures ranging from 1 to 10 bar. After plotting the results, the minimum cost appeared to occur between 4 and 5 bar.

To refine the analysis in this region, additional data points at 4.25, 4.5, and 4.75 bar were computed and added to the dataset.

An Excel polynomial trendline of sixth degree was then fitted to the cost versus pressure data. The resulting polynomial equation was extracted and entered into a dedicated cell in the spreadsheet, with the HP discharge pressure treated as a variable labeled *Optimal Pressure*.

Using this polynomial cost function, Excel's Solver tool was applied to determine the pressure that minimizes total annual operating cost. Solver was configured to minimize the cost cell by varying the *Optimal Pressure* variable. Figure 10 presents the annual operating cost as a function of HP turbine discharge pressure. Using Excel's Solver tool to minimize the polynomial cost function, the optimal operating point was determined to be:

$$P_{\text{HP,opt}} = 4.896 \text{ bar}$$

with a corresponding minimum annual operating cost of:

$$\$222,458.38 \text{ per operating year}$$

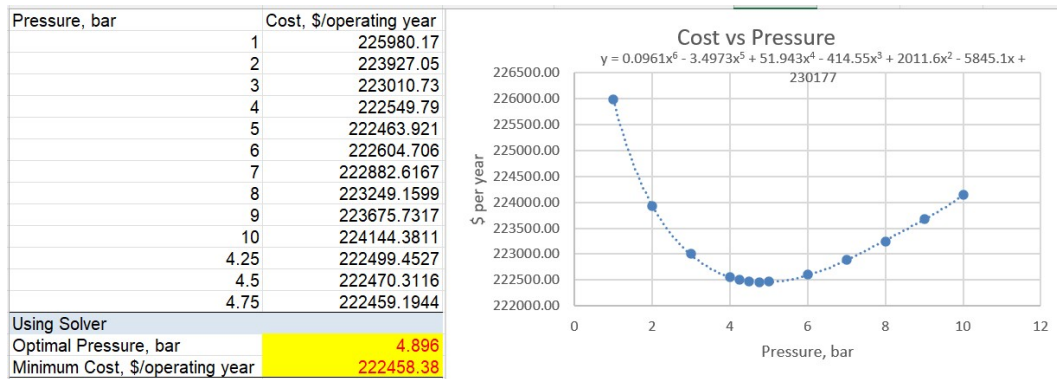


Figure 10: Cost vs Pressure optimization

11 Natural Gas Flow Rate (Q11)

To determine the mass flow rate of natural gas, the heat supplied to the boiler must first be adjusted to account for boiler efficiency. The heat transferred to the working fluid is denoted as $\dot{Q}_{in}$ (kW).

Because the boiler efficiency is 85%, the required fuel heat input is:

$$\dot{Q}_{fuel} = \frac{\dot{Q}_{in}}{0.85}$$

Since 1 kW = 1 kJ/s, the fuel heat rate can be converted to MJ/s by:

$$\dot{Q}_{fuel} \left(\frac{1 \text{ MJ}}{1000 \text{ kJ}} \right)$$

The mass flow rate of natural gas is then obtained by dividing the fuel heat rate by the higher heating value (HHV) of methane:

$$\dot{m}_{NG} = \frac{\dot{Q}_{fuel}}{\text{HHV}}$$

Substituting the efficiency relation gives:

$$\dot{m}_{NG} = \frac{\dot{Q}_{in}}{0.85 \times \text{HHV}}$$

where HHV is expressed in MJ/kg and $\dot{Q}_{in}$ is expressed in MJ/s, resulting in $\dot{m}_{NG}$ in kg/s.

The calculated natural gas mass flow rate (kg/s) was converted to a molar flow rate by dividing by the molar mass of methane:

$$\dot{n}_{NG} = \frac{\dot{m}_{NG}}{M_{CH_4}}$$

$$\dot{n}_{NG} = 0.0019 \text{ kmol/s}$$

where M_{CH_4} is the molar mass of methane (kg/kmol).

The resulting molar flow rate (kmol/s) was then converted to million standard cubic feet per day (MMSCFD) using the calculator's built-in conversion function:

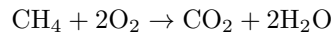
=KMOLperSECToMMSCFD(NG_MolarFlow_Cell)

This yields the required natural gas volumetric flow rate of:

$$\dot{V}_{NG} = 0.1407 \text{ MMSCFD}$$

12 Combustion Model Setup (Q12)

The boiler fuel is assumed to be 100% methane and the combustion reaction is:



A control volume is drawn around the burner/boiler combustion system with the following inlet streams:

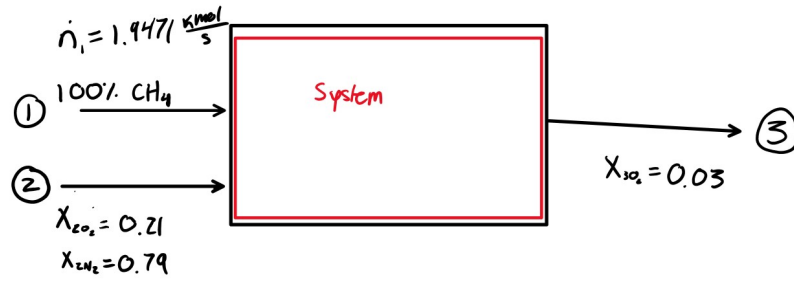


Figure 11: Diagram of the system under investigation. The molar flow rate for stream one is .0019 kmol/s, it is incorrect in the figure

- Stream 1: methane (100% CH₄) with molar flow rate $\dot{n}_1$
 - Stream 2: air with $x_{O_2,2} = 0.21$ and $x_{N_2,2} = 0.79$ and molar flow rate $\dot{n}_2$
- and the outlet stream:
- Stream 3: dry flue gas with oxygen mole fraction $x_{O_2,3} = 0.03$ and molar flow rate $\dot{n}_3$

Methane Balance and Extent of Reaction

Let ξ be the extent of reaction (kmol/s). A component balance on methane gives:

$$\dot{n}_{CH_4,3} = \dot{n}_{CH_4,1} - \xi$$

Because complete combustion is assumed, methane is fully consumed:

$$\dot{n}_{CH_4,3} = 0$$

Therefore,

$$\xi = \dot{n}_{CH_4,1} = \dot{n}_1 = 0.0019 \text{ kmol/s}$$

Oxygen Balance

A component balance on oxygen yields:

$$\dot{n}_{O_2,3} = \dot{n}_{O_2,2} - 2\xi$$

Using mole fractions to express oxygen molar flow rates:

$$x_{O_2,3} \dot{n}_3 = x_{O_2,2} \dot{n}_2 - 2\xi$$

With the specified compositions $x_{O_2,3} = 0.03$ and $x_{O_2,2} = 0.21$, this becomes:

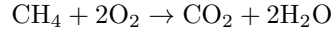
$$0.03 \dot{n}_3 = 0.21 \dot{n}_2 - 2\xi$$

where $\xi = \dot{n}_1$ from the methane balance:

$$0.03 \dot{n}_3 = 0.21 \dot{n}_2 - 2\dot{n}_1 \quad (1)$$

Carbon Dioxide Balance

A component balance on carbon dioxide is performed next. Because there is no CO_2 in either inlet stream, the outlet CO_2 molar flow rate is generated entirely by the reaction. From the stoichiometry,



the stoichiometric coefficient of CO_2 is 1, therefore the CO_2 molar flow rate in the outlet stream is equal to the extent of reaction:

$$\dot{n}_{\text{CO}_2,3} = \xi$$

In Stream 3, the CO_2 molar flow rate may also be written using its mole fraction:

$$\dot{n}_{\text{CO}_2,3} = x_{\text{CO}_2,3} \dot{n}_3$$

Combining these relations gives:

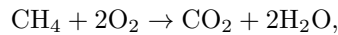
$$x_{\text{CO}_2,3} \dot{n}_3 = \xi \quad (2)$$

Flue Gas Mole Fractions and Nitrogen Balance

For the outlet (dry) flue gas stream (Stream 3), the mole fractions must satisfy:

$$x_{O_2,3} + x_{\text{CO}_2,3} + x_{\text{H}_2\text{O},3} + x_{N_2,3} = 1$$

Let $x_{\text{CO}_2,3}$ remain as an unknown. From the reaction stoichiometry,



water is produced at twice the molar rate of CO_2 . Under the same basis for Stream 3, this gives the relationship:

$$x_{\text{H}_2\text{O},3} = 2x_{\text{CO}_2,3}$$

The oxygen mole fraction is specified as $x_{O_2,3} = 0.03$. Substituting these relations into the mole fraction summation yields:

$$x_{N_2,3} = 1 - x_{O_2,3} - x_{CO_2,3} - x_{H_2O,3} = 1 - x_{O_2,3} - 3x_{CO_2,3}$$

Nitrogen Balance

Nitrogen is treated as inert and therefore does not participate in the reaction. A component balance on nitrogen gives:

$$\dot{n}_{N_2,2} = \dot{n}_{N_2,3}$$

Expressing each term in mole-fraction form:

$$x_{N_2,2} \dot{n}_2 = x_{N_2,3} \dot{n}_3 \quad (3)$$

where $x_{N_2,2} = 0.79$ for air, and $x_{N_2,3}$ is given by:

$$x_{N_2,3} = 1 - x_{O_2,3} - 3x_{CO_2,3}$$

Substituting the expression for $x_{N_2,3}$ into Equation (3), we obtain:

$$x_{N_2,2} \dot{n}_2 = (1 - x_{O_2,3} - 3x_{CO_2,3}) \dot{n}_3 \quad (4)$$

Eliminating Unknown Mole Fractions and Solving for Air Flow

From the CO_2 relation in Equation (2),

$$x_{CO_2,3} \dot{n}_3 = \xi$$

the CO_2 mole fraction can be isolated as:

$$x_{CO_2,3} = \frac{\xi}{\dot{n}_3} \quad (5)$$

Substituting Equation (5) into the nitrogen balance form (Equation (4)),

$$0.79 \dot{n}_2 = (1 - 0.03 - 3x_{CO_2,3}) \dot{n}_3,$$

gives:

$$0.79 \dot{n}_2 = \left(1 - 0.03 - 3 \frac{\xi}{\dot{n}_3}\right) \dot{n}_3 \quad (6)$$

Distributing $\dot{n}_3$ yields the simplified form:

$$0.79 \dot{n}_2 = (0.97) \dot{n}_3 - 3\xi \quad (7)$$

Next, the oxygen balance previously developed is:

$$0.03 \dot{n}_3 = 0.21 \dot{n}_2 - 2\xi \quad (8)$$

Equation (8) can be rearranged to isolate the total molar flow rate of Stream 3:

$$\dot{n}_3 = \frac{0.21 \dot{n}_2 - 2\xi}{0.03} \quad (9)$$

Finally, substituting Equation (9) into Equation (7) produces an expression in which $\dot{n}_2$ is the only remaining unknown:

$$0.79 \dot{n}_2 = 0.97 \left(\frac{0.21 \dot{n}_2 - 2\xi}{0.03} \right) - 3\xi \quad (10)$$

Because the extent of reaction is known from the methane balance ($\xi = \dot{n}_1$), Equation (10) contains only one unknown, $\dot{n}_2$, which can then be solved directly.

The previously determined extent of reaction ξ was substituted directly and the resulting expression was written in residual form. Specifically, after substituting the known value of ξ , Equation(10) was rearranged by subtracting the left-hand side from the right-hand side to obtain an equation equal to zero:

$$0 = 0.97 \left(\frac{0.21 \dot{n}_2 - 2\xi}{0.03} \right) - 3\xi - 0.79 \dot{n}_2 \quad (11)$$

Excel implementation. Equation (11) was implemented in a dedicated cell using references to: (i) the cell containing the known extent of reaction ξ , and (ii) a cell containing a guess for the air molar flow rate $\dot{n}_2$.

Excel's Goal Seek was then used to set the residual cell (Equation (11)) equal to zero by changing the guessed value of $\dot{n}_2$. The converged value of $\dot{n}_2$ corresponds to the required air molar flow rate satisfying the combined O₂ and N₂ balances.

Applying Goal Seek to Equation (11) yields the required air molar flow rate:

$$\dot{n}_2 = 21.9594 \text{ kmol/s}$$

This molar flow rate was then converted to million standard cubic feet per day (MMSCFD) using the calculator's built-in conversion function:

=KMOLperSECToMMSCFD(Stream2_MolarFlow_Cell)

The resulting required air flow rate is:

$\dot{V}_{air} = 1.5873 \text{ MMSCFD}$

13 CO₂ Emissions Calculation(Q9/Q10)

the stoichiometric coefficients of CH₄ and CO₂ are 1:1. Therefore, the molar flow rate of CO₂ produced equals the molar flow rate of methane fed:

$$\dot{n}_{CO_2} = \dot{n}_{CH_4} = \xi$$

The corresponding CO₂ mass flow rate is obtained using the molar mass of CO₂:

$$\dot{m}_{CO_2} = \dot{n}_{CO_2} M_{CO_2}$$

This mass flow rate (kg/s) was converted to lb/s using the spreadsheet conversion function:

$$=KGtoLB(CO2_MassFlow_kgps_Cell)$$

The CO₂ emissions were then converted to an annual basis:

$$\dot{m}_{CO_2} \text{ (lb/s)} \times 3600 \frac{\text{s}}{\text{hr}} = \dot{m}_{CO_2} \text{ (lb/hr)}$$

$$\dot{m}_{CO_2} \text{ (lb/hr)} \times (\text{operating hr/yr}) = m_{CO_2} \text{ (lb/yr)}$$

Finally, pounds were converted to tons and then to gigatons:

$$m_{CO_2} \text{ (tons/yr)} = \frac{m_{CO_2} \text{ (lb/yr)}}{2000}$$

$$m_{CO_2} \text{ (Gt/yr)} = m_{CO_2} \text{ (tons/yr)} \times 10^{-9}$$

Using fully referenced Excel calculations and consistent unit conversions, the annual CO₂ emissions are determined to be:

0.00000272 Gt CO ₂ /operating year

According to data reported on <https://ourworldindata.org/co2-emissions>, global CO₂ emissions from fossil fuels and industry in the year 2024 were approximately 38.60 Gt per year. Comparing the plant's annual CO₂ emissions to this global total:

$$\text{Plant Contribution} = \frac{0.00000272 \text{ Gt CO}_2/\text{yr}}{38.60 \text{ Gt CO}_2/\text{yr}}$$

Using Excel's percentage function, this corresponds to a contribution of approximately:

0.000007%

of worldwide CO₂ emissions in 2024.

14 NO_x Emissions per Operating Year (Q13)

The NO_x emissions were computed using the provided emission factor.

$$EF_{NO_x} = \frac{\text{lb NO}_x}{\text{MMBtu}}$$

where EF_{NO_x} is stored in cell F58. The operating time is t_{op} (cell F50, hr/operating-year). The fuel heat-release rate from methane combustion is computed in the spreadsheet as $\dot{Q}_{\text{fuel}}$ (cell O33/F42, kW).

Annual NO_x emissions are calculated as:

$$m_{NO_x, yr} = EF_{NO_x} \dot{Q}_{\text{fuel}} (\text{MMBtu/hr}) t_{op}$$

Because $\dot{Q}_{\text{fuel}}$ is available in kW, it must be converted to MMBtu/hr. The CHE calculator function `MMBTUperHRtoKW()` converts MMBtu/hr to kW; therefore, its inverse is used to convert kW to MMBtu/hr. Finally, the CHE calculator conversion `lbtokg()` is applied to report the result in kg per operating year.

To compute the annual NO_x emissions, the following Excel cell expression was implemented directly in the worksheet:

```
=lbtokg((MMBTUperHRtoKW(1/F58/F50)/(033/F42))^-1)
```

Evaluating this cell in Excel gives the total NO_x emissions of:

1073.84 kg NO_x /operating year

Conclusion

This assignment evaluated the performance of a 500 kW regenerative Rankine cycle designed for power generation at O'Bleness Hospital. Unlike the ideal Rankine cycle, the proposed configuration incorporates steam extraction and an open feedwater heater to improve thermal performance and reduce irreversibilities.

A complete thermodynamic, economic, and environmental analysis was performed. Energy and entropy balances were applied to each unit operation, followed by an availability analysis to quantify lost work. Economic optimization was conducted by varying the HP turbine discharge pressure, and environmental impact was assessed through CO_2 and NO_x emission calculations.

The regenerative configuration achieved a thermal efficiency of 29% and a thermodynamic efficiency of 38%, compared to a maximum theoretical efficiency of 76% based on the 1250 K heat source and 300 K sink temperatures. The total lost work of 820.29 kW reflects the impact of system irreversibilities, particularly within the boiler and turbine stages.

Cost optimization yielded a minimum annual operating cost of \$222,458.38 at an optimal HP turbine discharge pressure of 4.896 bar, demonstrating the economic value of systematic parameter optimization.

From an environmental standpoint, the plant emits 0.0000027 Gt of CO_2 per operating year, corresponding to approximately 0.000007% of global CO_2 emissions (38.60 Gt/year in 2024). NO_x emissions were calculated to be 1073.84 kg per operating year. While modest in global comparison, these values highlight the importance of efficiency improvements in reducing fuel consumption and associated emissions.

Overall, this analysis demonstrates that regenerative cycle enhancements provide meaningful improvements over the ideal Rankine cycle by increasing efficiency, lowering operating cost, and reducing environmental impact. Such evaluations are critical in the design of modern power systems, where thermodynamic performance, economic feasibility, and sustainability must be simultaneously considered.

Table 1: Summary of Final Results

Parameter	Value
Steam Flow Rate	2.0923 MMSCFD
Mass Fraction of Bleed Steam (S3)	0.1295
Thermal Efficiency	29%
Thermodynamic Efficiency	38%
Total Lost Work	820.29 kW
Maximum Theoretical Efficiency	76%
Minimum Annual Cost of Utilities	\$222,458.38 / year
CO ₂ Emissions	0.0000027 Gt / year
Percent of Global CO ₂ Emissions	0.000007%
NO _x Emissions	1073.84 kg / year

References

- [1] M. Caracotsios, *An Introduction to Chemical Engineering Thermodynamics: Theory and Applications*, First Edition, course textbook, Chemical Engineering Department, University of Illinois Chicago, 2026.
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- [3] OpenAI, “ChatGPT,” Large Language Model, 2026. [Online]. Available: <https://chat.openai.com>. [Accessed: Feb. 2026].